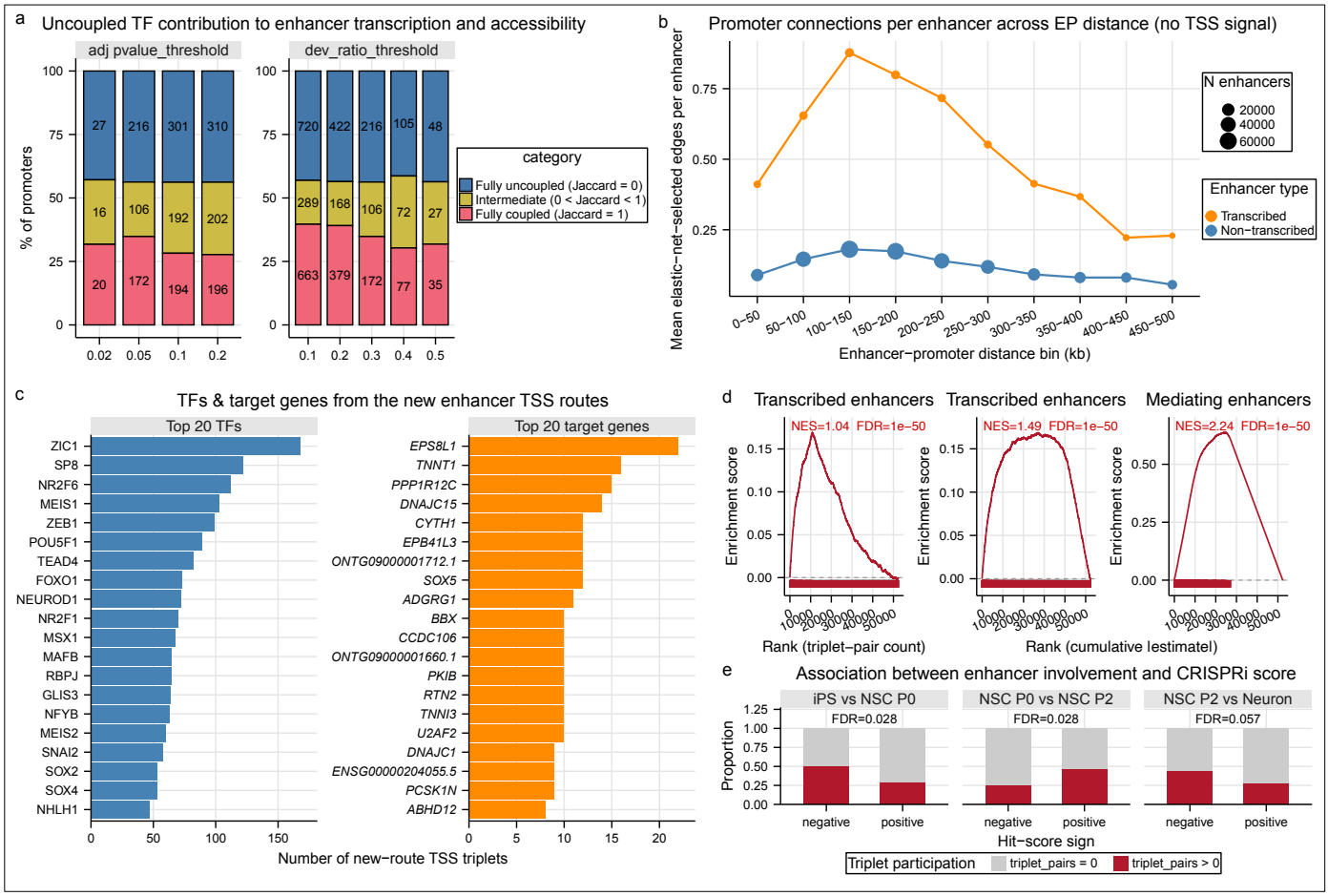


Extended Figure 6 | Robustness of enhancer regulatory decoupling and network connectivity.

a, Overlap between TF sets associated with enhancer accessibility and transcription across adjusted p-value thresholds. Only transcribed enhancers with at least one TF association in both modalities are included. Enhancers are grouped as completely distinct TF sets across accessibility and transcription modalities, partially overlapping or completely overlapping. **b**, As in **a**, across deviance-ratio thresholds. The deviance-ratio threshold of 0.3 is used for the final GRN. **c**, Complementary to **Fig. 6b** using a control GRN reconstructed without enhancer TSS predictors, where transcribed and non-transcribed enhancer are represented only by accessibility measurements to provide an equal-footing comparison. Mean number of elastic-net-selected enhancer–promoter edges per enhancer across genomic-distance bins, stratified by transcribed (orange) and non-transcribed (blue) enhancers. Dot size represents the number of enhancers. **d**, Distribution of TFs and target genes from the TF-enhancer-promoter regulatory routes uniquely recovered by incorporating enhancer transcription signal. Bars show the 20 TFs contributing the largest numbers of new routes (left) and the 20 target-gene promoters receiving the largest numbers of new routes (right). **e**, Gene set enrichment analysis testing whether transcribed enhancers (left) or mediating enhancers (right) are enriched toward higher enhancer centrality, quantified by triplet-pair count or cumulative absolute regulatory weight, among all 52,334 enhancers. Enhancers were ranked from highest to lowest centrality, with membership indicated below each enrichment profile. **f**, Association between participation in TF-enhancer-promoter triplet (at least one) and the direction of CRISPRi perturbation phenotypes. For each indicated differentiation contrast, screen-scored enhancers were classified by triplet participation and positive or negative hit score, and associations were assessed using Fisher's exact test.